Information Quality

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What is Information Quality?

- Many Definitions
- -Accuracy
- -Fitness for use
- Accurate, complete, and timely : information that is free from defect
- Right information at the right time, at the right place, for the right people
- The application of Total Quality Management principles to building Information Products

Definition of IQ

- Information Quality is an Information Science **discipline** concerned with:
- **Maximizing the value and minimizing the risk of an organization's information assets.**
- **Assuring that the information products produced by the organization will create value for the customers who use them.**

IQ Knowledge & Practice

- Information Quality is a discipline, a body of knowledge and practice covering :
  - Information quality value & business impact
  - Information quality strategy & governance
  - Information quality environment & culture
  - Information quality measurement & improvement
  - Sustaining information quality
  - Information architecture quality(six domains of IQ)

Data Quality

Why Study Information Quality?

- It is important!

  Poor IQ:

  - Cost billions of dollars each year
  - Costs lives

- It is a new and quickly growing field!

  - *Organizations are embracing data and information as important assets*

- YOU can make a difference!

We Are a Community

- More than just students
  - We support each other
  - We respect each other especially

    professors, teaching assistants, senior class members, graduates
- Take advantage to
  - Build your network
  - Build your professional credentials
  - Gain professional experience

Course Description

- Provides a rigorous exploration of information quality concepts, assessment, and problems in organizational information systems, databases and data warehouses.
- A combination of literature review and hands-on projects is used to develop knowledge and ability to meet objectives.

Course Objectives

Give students an understanding of

- Definition of IQ
- Basic IQ terminology and concepts
- Principles of IQ study and practice
- Economic and human cost of poor IQ
- Best practices for improving IQ
- Role of statistics in IQ

Course Mission

- The course designed to meet the growing demands by government and industry for qualified professionals with a undergraduate-level degree. After finishing the study, you can get:
- 1) Concepts, principles, tools, and models essential in defining, measuring, analyzing, and improving information quality

Course Mission

- 2) Information quality policies, strategies, programs, and standards (e.g. ISO 8000)
- 3) The role of information quality in operational improvement, data privacy and protection, risk management, data governance, information architecture, entity resolution, and master data management

Books & Resources

- *Data Quality*

Nie Yu,

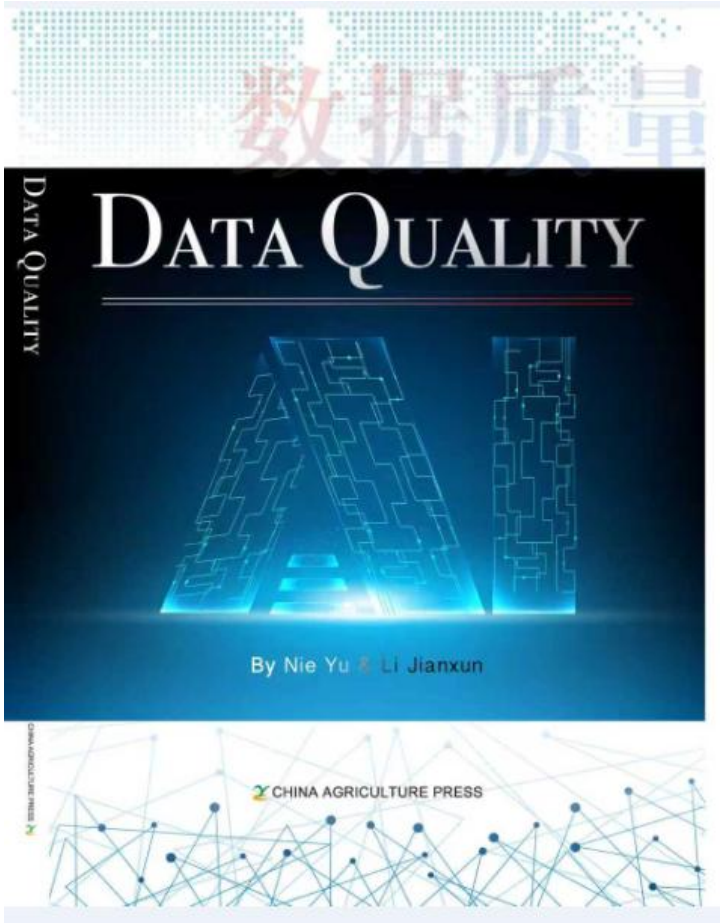
Li Jianxun,

China

Agriculture

Press

2026



Books & Resources

- *Journey to Data Quality*,

  Lee, Lauria, Chengalur-Smith, Wang, MIT Press © 2006

- *Measuring Data Quality for Ongoing Improvement*,
- Laura Sebastian Coleman, Morgan Kaufmann, © 2013
- **大数据安全：技术与管理，**
- 王瑞民，2021。

Graded Work

- Exam
- Written assignments
- Laboratory Exercises

THANK YOU!